

React Assignment

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Part A • Warm-up: quick concepts

1. In your own words: what is a component in React?

Ans. A React component is a reusable, independent piece of UI that contains its own structure and logic. Components help build complex applications by combining smaller, manageable parts.

2. This JSX is broken: `<div class="card">`what is wrong, and what is the fix?

Ans. JSX uses `className` instead of `class` because `class` is a reserved keyword in JavaScript.

3. Props vs state: which one can a component change itself, and which one is read-only?

Ans. State can be changed by the component itself using state update functions. Props are read-only and are passed into the component by its parent.

4. What does the empty array do in `useEffect(() => { ... }, [])`?

Ans. The empty array `[]` makes the `useEffect` run only once after the component's first render (mount). Without it, the effect runs after every render.

5. Why does React need a key prop when you render a list with `.map()` ?

Ans. React uses a key prop to uniquely identify each item in a list so it can efficiently track changes, additions, and removals.

Part B • Bug hunt

BUG 1 The alert fires as soon as the page loads — not when the button is clicked.

```
Function SaveButton(){  
  
  Function handleSave(){  
  
    Alert('Saved! ');  
  
  }  
  
  Return<button onClick={handleSave()}>Save</button>;  
  
}
```

Ans. The bug is at `onClick={handleSave()}`, `handleSave()` is being called immediately during rendering, so `alert('Saved!')` runs as soon as the component loads

BUG 2 Clicking “Add” updates the array — but the screen never changes.

```
function TodoApp() {  
  
  const [todos, setTodos] = useState(['Learn JSX']);  
  
  function addTodo() {  
  
    todos.push('New task');  
  
    setTodos(todos);  
  
  }  
  
  return <button onClick={addTodo}>Add</button>;  
  
}
```

Ans. The bug is at `todos.push()` mutates the existing state array directly. React expects state to be treated as immutable. React may not detect a change and may not re-render the component.

BUG 3 The list renders, but the console shows a warning on every refresh.

```
const skills = [  
  
  { id: 1, name: 'HTML' },  
  
  { id: 2, name: 'CSS' },  
  
  { id: 3, name: 'JavaScript' },  
  
];  
  
function SkillList() {  
  
  return (  
  
    <ul>  
  
      {skills.map(skill => (  
  
        <li>{skill.name}</li>  
  
      ))}  
  
    </ul>  
  
  );  
  
}
```

Ans. The bug is at `{skills.map(skill => (`

```
<li>{skill.name}</li>
```

```
}}
```

Each item rendered by `map()` should have a unique `key` prop. Without it, React shows a warning and may not efficiently update the list.

Profile github link- [yashilavaniacg/React_CGU](https://github.com/yashilavaniacg/React_CGU)